OUR SOLUTION

DrugNova is an AI powered drug repurposing platform that helps researchers identify potential new uses for existing drugs by analyzing relationships between diseases, genes, proteins, biological pathways, and drugs. The platform combines biomedical databases, a knowledge graph, protein-target analysis, explainable AI, and a machine learning prediction model to discover promising drug candidates faster and at a lower cost than traditional drug discovery methods.

CORE COMPONENTS

1. Biomedical Data Integration
   - Collects and unifies data from DrugBank, ChEMBL, DisGeNET, Open Targets, BindingDB, and PubChem.
2. Knowledge Graph Engine
   - Creates connections between diseases, genes, proteins, pathways, and drugs to uncover hidden relationships.
3. Protein Target Analysis
   - Identifies key proteins involved in a disease and analyzes potential drug interactions with those targets.
4. Drug Repurposing Recommendation Engine
   - Suggests existing drugs that may be effective for treating new diseases based on biological and molecular similarities.
5. Machine Learning Prediction Model
   - Trained on historical drug-target, drug-disease, and protein interaction datasets to predict the likelihood of a successful drug repurposing opportunity.
6. Explainable AI Module
   - Provides reasoning behind every recommendation, showing why a specific drug was suggested.
7. Interactive Visualization Dashboard
   - Visualizes disease-drug-protein relationships and molecular interaction networks for researchers.

TARGET USERS

- Pharmaceutical Companies
- Biotechnology Startups
- Biomedical Researchers
- Drug Discovery Scientists
- Academic & Research Institutions
- Healthcare Innovators
- Clinical Researchers

TECH STACK

Frontend

- React.js
- Material UI (MUI)
- Three.js (3D Molecular Visualization)
- Recharts / D3.js

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

AI/ML

- Python
- Scikit learn
- PyTorch
- NetworkX
- Graph Neural Networks (Future)

APIs & Data Sources

- DrugBank
- ChEMBL
- DisGeNET
- BindingDB
- PubChem
- UniProt

ML MODEL (ADVANCED COMPONENT)

- **Input:** Disease features, protein targets, pathway data, drug properties, and known interactions.
- **Output:** Drug repurposing probability score and ranked candidate drugs.
- **Potential Models:** Graph Neural Networks (GNNs), XGBoost, Random Forest, or Deep Learning models trained on biomedical interaction datasets.

EXPECTED IMPACT

- Reduce the time required to identify potential treatments.
- Lower early stage drug discovery and research costs.
- Increase utilization of existing approved drugs.
- Help researchers uncover hidden disease drug relationships.
- Accelerate therapeutic discovery for rare and common diseases.
- Improve decision making through explainable AI insights.
- Enable faster hypothesis generation and validation.

KEY DIFFERENTIATORS

- AI-driven **drug repurposing** for faster treatment discovery.
- **Knowledge graph** linking diseases, genes, proteins, and drugs.
- **Explainable AI** with transparent recommendation reasoning.
- Interactive **3D protein-drug visualization and simulation**.
- **Researcher-controlled molecular experimentation**, not just AI predictions.
- **End-to-end platform** for discovery, analysis, and validation.

POSSIBLE REVENUE STREAMS

Freemium Model – Free basic access with premium features behind a subscription.

Individual Researcher Subscriptions – Monthly or annual premium plans.

Academic & Institutional Plans – Subscription packages for universities and research labs.

Enterprise Licensing – Premium access for pharmaceutical and biotech companies.

Advertisements – Relevant ads from healthcare, biotech, and research-related organizations.

Premium Reports & Analytics – Paid access to detailed insights and candidate evaluations.